

23SLN7

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Problem

Let a, b, c, d be positive integers satisfying

$$\frac{ab}{a+b} + \frac{cd}{c+d} = \frac{(a+b)(c+d)}{a+b+c+d}.$$

Determine all possible values of $a+b+c+d$.

Solution. The answer is all non-squarefree positive integers.

We have $ab(c+d) + cd(a+b) = \frac{(a+b)^2(c+d)^2}{a+b+c+d}$. Let $a+b = x$ and $c+d = y$. Then, $s = x+y \mid x^2y^2$.

Consider any prime p dividing s . Then, $p \mid s \mid x^2y^2$. If $p \mid x$, then $p \mid s \mid x+y$ implies that $p \mid y$. Similarly, if $p \mid y$, then $p \mid s \mid x+y$ implies that $p \mid x$.

Let $\{p_1, p_2, \dots, p_k\}$ be the set of primes dividing s (it is trivial that $s > 1$). Observe that s the least possible such number. Since we have that if $p \mid x$ then $p \mid y$, this implies that $\{p_1, p_2, \dots, p_k\}$ is the set of primes dividing x and y .

Since we have $0 < x, y < s$, this contradicts the minimality.

For construction, consider any non-squarefree positive integer of the form mn^2 . Then, $(a, b, c, d) = (m(n-1)^2, m, m(n-1), m(n-1))$ works.